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GAL SASSON LAZOVSKY
TUVAL RAZ
YOED N. KENETT

# The Art of Creative Inquiry—From Question Asking to Prompt Engineering

ABSTRACT

As artificial intelligence and natural language processing methods rapidly develop, communication plays a pivotal role in every-day interactions. In this theoretical paper, we explore the overlap and commonalities between question-asking and prompt engineering. While seemingly distinct, these processes share a common foundation in essential skills like creativity, critical thinking, and cognitive flexibility. We contend that prompt engineering, the art of crafting cues for language models, and question-asking, the skill of formulating inquiries, form a symbiotic relationship. Delving into question complexity through Bloom's taxonomy and diverse types of questions, we propose strategies for not only efficient but also engaging prompt design. Our theoretical contribution emphasizes the dynamic role of creativity in both processes, offering intriguing perspectives on human–machine interactions and advancing our understanding of language models and communication skills.

*Keywords:* prompt engineering, question asking, creativity, large language models.

Co-creativity is generally defined as multiple parties contributing to a creative process in a blended manner (Davis, 2013). Although in some creative situations tasks can be completed through a distribution of labor, the results in this case only represent the sum of each individual's contribution. Co-creativity goes beyond this labor allocation model and allows individuals to contribute in a collaborative and synthetic manner. Thus, ideas can be fused, combined, and added onto each other such that they originate from a myriad of personalities and team members' motivations (Candy & Edmonds, 2002). It can then yield more creative solutions than if each party completed each task separately and added them together, that is, the sum is greater than its parts. Co-creativity is becoming especially relevant now in the age of generative artificial intelligence (AI) where highly advanced generative AI can now begin to match humans in performance on creativity tests (Hubert, Awa, & Zabelina, 2024) and has the potential to augment the creativity of every knowledge worker on Earth (Rafner et al., 2023; Rafner, Beaty, Kaufman, Lubart, & Sherson, 2023). Although autonomously operating AI agents will certainly continue to amaze us, such technologies in the hands of creative experts can offer potentially endless avenues of innovative abilities.

An integral part of successful co-creativity is good communication (Kantosalo & Takala, 2020). Decades of human–computer interaction research on conversational interfaces suggest that people's perceived utility of these technologies can be shaped by how these agents communicate and clarify their capabilities (Liao & Vaughan, 2023). Contemporary AI models such as ChatGPT and GPT-4 (Achiam et al., 2024) are becoming increasingly popular tools in many linguistic human–computer interactions. These large language models (LLM) are a class of deep neural networks that undergo pre-training on large amounts of text data, for the purpose of understanding and generating language, and can then aid in machine translation, question answering, automatic summarization, and other language tasks (Desai, Sharma, & Saha, 2023). Thus, in the ever-evolving landscape of AI and natural language processing, the art of clearly communicating, which includes how the models introduce and clarify their capabilities, take initiatives, repair errors, and respond to chat requests, stands at the forefront of potential groundbreaking advancements in the way we interact with these systems.

While question-asking stands as a fundamental process of shaping communications between humans (Huang, Yeomans, Brooks, Minson, & Gino, 2017), prompt engineering has emerged as a key pillar in shaping the interactions between humans and language models (Meskó, 2023; White et al., 2023).



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Prompt-engineering involves crafting precise cues to elicit specific responses from language models (Heston & Khun, 2023), while question-asking delves into the skill of formulating inquiries that reduce entropy and uncertainty and stimulate critical thinking and meaningful discussions (Flammer, 1981). Despite their distinct roles, a profound connection lies at the heart of these practices—they require remarkably similar capabilities.

Here, we will present some of the existing theories about question-asking, prompt engineering, and their interrelationship. Then, we will review some of the shared skills and intricacies that intertwine these essential processes and discuss how theories from one process could be applied to the other to further understand the relationship between them. Through a multidimensional lens, we will shed light on how honing these capabilities transforms not only our interactions with language models but also our understanding of human cognition and the very nature of inquiry itself. Finally, we will argue that to engage in successful co-creativity with LLMs, we need to be able to use better prompts based on more complex and tailored questions which fit the goal of our inquiry. After taking a deep dive into this exploration, we conclude that understanding each of these processes will help us improve our understanding of human communication skills and language models, all of whom are here to stay.

## QUESTION-ASKING: RELATED WORK AND EXISTING THEORIES

Question-asking is a fundamental tool for gaining information, requesting clarification, or understanding a particular topic or subject (Flammer, 1981; Rothe, Lake, & Gureckis, 2018). It is a concept that reflects aspects of human curiosity and the pursuit of knowledge and stands as the basis of communication and learning (Ivancovsky, Baror, & Bar, 2023; Jirout, 2011; Kedrick, Schrater, & Koutstaal, 2023; Kenett, Humphries, & Chatterjee, 2023).

From our earliest stages of cognitive development, we instinctively inquire about the world around us, seeking to understand its complexities and uncover its mysteries (Herwig, 1982; Ruggeri, Xu, & Lombrozo, 2019). In everyday life, people constantly ask questions to interact with others, make informed decisions, and navigate the world around them. It is through questioning that we can solve problems and adapt to new situations, making it an essential skill for personal and professional development. The willingness to ask questions, even about widely accepted beliefs, opens the door to progress and innovation, leading to groundbreaking discoveries in various fields (Poirier et al., 2017). Question-asking has been shown to be linked to the creative process (Acar, Berthiaume, & Johnson, 2023; Raz, Reiter-Palmon, & Kenett, 2023), and research has shown it to be a valid measure of creativity (Kenett et al., 2023; Yager, 1996). Question-asking has also been hypothesized as a critical mechanism in information seeking behavior (Raz & Kenett, 2024), likely by facilitating problem finding, the first stage of the creative process (Raz et al., 2023; Reiter-Palmon & Robinson, 2009). When developing questions, the asker must articulate and express the problem or perplexity they encounter and transition from perplexity to the formulation of the problem and expression of a question (Van Der Meij, 1994). Despite its importance in understanding human cognition, learning, and creativity, scarce research has been made regarding the cognitive mechanisms required to ask questions (Raz et al., 2023; Raz & Kenett, 2024; Sasson & Kenett, 2023).

Assessing the quality and effectiveness of questions poses a significant challenge, as it requires a nuanced understanding of the complexity inherent in each inquiry. The complexity of a question often reflects the depth of thought, creativity, and understanding of the asker, indicating a higher level of cognitive engagement with the subject matter. Complex questions tend to facilitate more profound discussions, drive deeper analysis, and encourage a broader exploration of ideas. However, evaluating this complexity involves considering the context in which questions are asked, the intent behind them, and the cognitive processes they engage, and thus determining what is a good question might have various considerations (Sasson & Kenett, 2023). In the upcoming sections, we will explore some of the suggested theories around question-asking strategies, processes, and assessments.

### BLOOM'S TAXONOMY: A HIERARCHICAL FRAMEWORK FOR LEARNING AND COGNITION

Questions can take various forms, ranging from simple queries to complex inquiries that delve into deeper complexities. They can be open-ended, allowing for broad exploration and discussion, or they can be closed-ended, seeking specific answers (Ortlieb et al., 2012).

Bloom, Krathwohl, and Masia (1956) proposed a taxonomy of question types, with increasing abstraction, ranging from the basic level of information retrieval questions to higher levels of synthesis and integration questions. The revised version of the taxonomy consists of six hierarchical levels, each representing a



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distinct level of cognitive complexity (Krathwohl, 2002): *remembering*—acquiring factual information from the long-term memory, including specific details, terms, or basic concepts, without needing to understand exactly what they mean; *understanding*—demonstrating the understanding of information by interpreting, explaining, and summarizing concepts or ideas; *applying*—using acquired knowledge to solve problems or apply the concepts in new situations; *analyzing*—examining information to break down complex concepts, identifying patterns, and analyzing relationships between various elements; *evaluating*—assessing and making judgment about information based on specified criteria while exhibiting critical thinking; and *creating*—integrating ideas, concepts, or information into something new and innovative. The Bloom taxonomy is thought to represent a cumulative hierarchy, where mastery of each simpler category is a prerequisite for mastery of the next more complex one (Krathwohl, 2002).

Recently, Raz et al. (2023) explored the relation between question-asking and creativity using Bloom's taxonomy to rate the complexity of questions asked for a creative question-asking task—the alternative questions task (AQT). The AQT requires participants to generate creative and unusual questions about common objects (e.g., pencil, book, shoe) such as "Who invented the first pencil?" or "What library has the most books?". Responses were rated separately in terms of their creativity, using a 1 (not at all creative) to 5 (very creative) scoring method (Runco & Mraz, 1992), and their complexity, according to Bloom's taxonomy, such that for example the question "Where is the pencil?" would be rated as Bloom level 1—remembering, while the question "Would it be possible to creative biodegradable pencils?" would be rated Bloom level 6—creating. Each object cue was rated by five different independent raters. The authors observed good inter-rater agreement (alpha Cronbach >0.7) for the subjectively rated Bloom taxonomy scores and moderate inter-rater agreement for subjectively rated creativity (alpha Cronbach >0.6). When evaluating the relation between complex questions and creativity, the authors found that question complexity and creativity were positively related: Questions that were higher on the Bloom taxonomy (i.e., more complex) were also scored as more creative, and those that were less complex were scored as less creative. The researchers provided the first empirical proof that question complexity and creativity are related, such that stronger creative abilities will accompany stronger question-asking abilities.

Thus, the work of Raz et al. (2023) provides evidence that the Bloom taxonomy can be used to assess question complexity in empirical question tasks such as the AQT and that creativity is related to gradual progression from basic knowledge acquisition to higher order thinking skills. Therefore, it is possible to see how improving question-asking skills through more complex questions may affect the development of problem-solving abilities, creativity, and overall mastery of complex cognitive thinking by encouraging usage of higher-level questions which have been shown to encourage creativity, deep understanding, and problem-solving (Nappi, 2017).

## UTILIZING CATEGORICAL KNOWLEDGE: CONSTRAINT-SEEKING VS. HYPOTHESIS-SCANNING QUESTIONS

Another classification for distinguishing types of questions was suggested by Mosher and Hornsby (1966) who categorized them based on categorical knowledge and understanding into constraint-seeking and hypothesis-scanning questions. Constraint-seeking questions are aimed to filter down possibilities by applying broad constraints that affect multiple options simultaneously. For example, asking, "Is it a type of food?" helps to quickly separate edible items from non-edible ones, effectively narrowing the scope of potential object possibilities. On the other hand, hypothesis-scanning questions are more targeted, focusing on verifying or refuting a single hypothesis about a specific item. An example of such a question might be "Is it an orange?", which seeks to confirm whether the specific hypothesis of "orange" is correct, without addressing or ruling out other possibilities. Additionally, pseudo-constraints questions are inquiries phrased like constraint-seeking questions while they rule out only one hypothesis. For example, "Is it something other than an orange?" might seem broad but actually focuses on excluding just one possibility. This can be useful if the range of potential answers is very limited. When all equally probable correct solutions are presented, constraint-seeking questions often provide at least as much information as hypothesis-scanning questions: By ruling out multiple hypotheses at each step of the search process, they reduce the number of questions needed to identify the solution (Mosher & Hornsby, 1966; Ruggeri, Sim, & Xu, 2017).

These types of questions are aligned with different searching strategies and thus may reflect how people gather information through question construction. Early work conducted by Stafford (1966) found that choice of strategy (constraint-seeking vs. hypothesis-scanning) is associated with creativity and intelligence.



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It was further found that constraint-seeking questions are positively related to high intelligence and high creativity, whereas low intelligence and low creativity are associated with hypothesis-scanning questions.

In a study on question-asking, Sasson and Kenett (2023) conducted an exploratory data analysis of the Akinator online game, which simulates a genie asking questions to guess a character the user is thinking of. The research revealed insights into question-asking strategies and discovered that the Akinator's question-asking process mirrored some aspects of human question-asking, such as the use of multiple question-asking strategies during the process. This includes not just reducing entropy but also considering a wider array of factors, much like the strategic use of constraint-seeking versus hypothesis-scanning questions. On the Akinator's side, these results shed light on the potential influence of gamification characteristics on question specificity and types. These insights not only enhance our comprehension of question-asking dynamics but also underscore the relevance of diverse questioning techniques for prompt engineering, illustrating how strategic question formulation can significantly improve the interaction quality and output relevance of language models. This paves the way for a deeper examination of prompt engineering techniques that leverage the nuanced understanding of question types and strategies to optimize language model responses, which could potentially be done through research based on digital gaming (Allen et al., 2023; Pedersen et al., 2023; Rafner et al., 2022; Rafner, Beaty, et al., 2023; Rafner, Wang, et al., 2023).

## PROMPT ENGINEERING: RELATED WORK AND EXISTING THEORIES

Conversational LLMs, such as ChatGPT (Brown et al., 2020), are a powerful class of AI models devoted to generating text through predictive distributions. LLMs have already aroused great interest in the scientific community in various domains and tasks (Jablonka et al., 2023), such as in the field of creativity where these models have been used in measuring task performance nearly instantaneously on tasks such as assessing creativity of metaphors (DiStefano, Patterson, & Beaty, 2024) and responses to problem-solving tasks (Luchini et al., 2023).

LLMs are trained on a huge corpus of documents from which they learn the distribution over generating new text. When LLMs are used in new tasks, they rely solely on their original training and do not use supplemental, task-specific data, a paradigm known as in-context learning. To receive the desired outputs, the models are controlled through user-provided natural language specifications—also known as the prompt of the task, which illustrate how to complete the task. To enhance the application of LLMs, prompts can be tuned and engineered to achieve better results. The art of prompt engineering is focused on crafting prompts that encourage more human-like interactions (Ouyang et al., 2022). Recent work has shown that models are sensitive to the prompts given to them (Webson & Pavlick, 2022) and that small changes to prompts can result in large performance improvements (Gao, Fisch, & Chen, 2021), while poorly designed prompts may cause irrelevant or nonsensical answers (Bozkurt & Sharma, 2023). Yet, crafting effective prompts can be challenging and even more so for non-AI experts (Lin, 2024; Zamfirescu-Pereira, Wong, Hartmann, & Yang, 2023).

In this context, it is crucial to acknowledge that effective prompt engineering not only refines task completion but may also offer a valuable avenue for augmenting human performance across various tasks (Dwivedi et al., 2023; Urban et al., 2023). By harnessing the capabilities of LLMs, prompt engineering becomes a strategic tool in enhancing the efficiency and quality of tasks performed by humans, marking a significant synergy between AI output and human cognition. They can support human users in achieving higher levels of creativity, problem-solving, and decision-making and enhance human capabilities in a wide range of activities. For instance, in creative domains, well-engineered prompts can inspire writers and artists with novel ideas or perspectives they might not have considered (Chen & Chan, 2023; Kirkpatrick, 2023; Rafner, Beaty, et al., 2023; Rafner, Wang, et al., 2023; Wan et al., 2023). Effective prompt engineering also requires correct usage of examples, which in some cases can be either beneficial or harmful. Gonen, Iyer, Blevins, Smith, and Zettlemoyer (2022) found that in experimental prompt engineering studies involving LLMs, prompts with lower perplexity (indicating less specificity) tended to perform better, but crucially, this relationship was observed specifically in scenarios where the answer to the question was already known. This suggests that while simplicity may generally enhance performance, the context of the question—and especially its familiarity to the model—plays a critical role.

### LEARNING WITH OR WITHOUT EXAMPLES: ZERO-SHOT VS. FEW-SHOT

It is often sufficient for prompts to contain only instructions for the desired task, and not examples, as this is all that is needed in many cases. In other words, the LLM is capable of performing the downstream



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task without any additional training required. This is traditionally known as a zero-shot setting, as there is zero training data for the task of interest. In some situations, however, it may be advantageous to include some examples in the prompt, which is called a few-shot learning method. These examples help specify the desired behavior and output and can help further guide the model in achieving it (Lin, 2024).

Few-shot learning has been shown to improve task accuracy (Brown et al., 2020) and even the ordering of the examples can influence the final results and impact the difference between state-of-the-art and random performance (Lu, Bartolo, Moore, Riedel, & Stenetorp, 2022; Zhao, Wallace, Feng, Klein, & Singh, 2021). The fact that examples help models learn better and faster is not surprising, as this is also true for humans. To some degree, these models may understand instructions in ways analogous to humans' understanding of task instructions (Webson & Pavlick, 2022). Their linguistic-heavy capabilities make LLMs especially suited for implementation in fields where language data have a central role, such as psychology, where this new technology has the potential to transform the field (Demszky et al., 2023).

## MACHINE PSYCHOLOGY: THE CONNECTION BETWEEN QUESTION-ASKING AND PROMPT-ENGINEERING

While there may not be a singular theory dedicated explicitly to the relationship between question-asking and prompt engineering, researchers and practitioners in the fields of natural language processing, communication, and cognitive sciences continuously explore the synergy between these practices. A great deal of research has been done into the complex "psychology" of machine learning models, and especially LLMs, to gain a deeper understanding of their capacity to grasp human-like concepts and to consider the far-reaching implications of these advancements on society (Hagendorff, 2023). On the other hand, the idea that LLMs can help us gain a better understanding of human behavior, while not trying to computationally realize human-like cognition, is widely held to be true as well (Hardy, Sucholutsky, Thompson, & Griffiths, 2023; van Rooij et al., 2023).

Through advances in AI, researchers show that cognitive-like processes have spontaneously emerged in language models. AI systems exhibit theory of mind-like abilities without them being explicitly engineered to display them, but as an emergent byproduct of their training process on other data (Kosinski, 2023). This integration has enhanced the quality and naturalness of language generation, bringing language models closer to simulating human-like understanding and communication. Conversely, these same emulations of human-like abilities may also insert human-like biases, with Binz and Schulz (2023) finding that GPT-3 often responded to cognitive tasks analogously to humans, while also showcasing human-like biases. They also note that GPT-3 solved many of these tasks reasonably well, despite being only taught to predict future word occurrences (Binz & Schulz, 2023). Stella, Hills, and Kenett (2023) take this even further and argue that GPT-like LLMs may even contain non-human biases such as myopic overconfidence (under sampling and overlooking data, failing to ask clarifying questions, and short-term maximizing while failing to account for long-term potential gains) and hallucinations (fabricating information without any apparent awareness of its source).

Furthermore, linguists are grappling with questions regarding the extent to which these language models truly grasp the intricacies of language structure, semantics, and pragmatics. The field of linguistics is undeniably impacted by the rapid advancement of powerful tools capable of producing outputs that closely resemble human-generated texts. Thus, these tools are reigniting long-standing debates in linguistics that have persisted for decades (Agüera y Arcas, 2022; Jiang, Xu, Araki, & Neubig, 2020).

Due to their training on a huge corpus of documents and text for the purpose of understanding and generating language, LLMs are used for a variety of tasks involving language data (Vaswani et al., 2017). These models have shown demonstrable success across all fields of psychology in part due to them being able to measure and administer tasks that are relevant to psychologists nearly instantaneously. Thus, LLMs open possibilities for scale and efficiency in psychological research and practice that were unimaginable in the past (Demszky et al., 2023).

For example, LLMs can be used to generate experimental stimuli (Laverghetta Jr. & Licato, 2023), model word learning across the lifespan (Portelance, Degen, & Frank, 2020), predict personality traits from text (Peters & Matz, 2023), and predict the creativity of metaphors (DiStefano et al., 2024) and responses to problem-solving tasks (Luchini et al., 2023). This wide variety of uses for LLMs can mainly be attributed to the rise of their emergent abilities—capabilities that large models gain only after being exposed to vast amounts of textual data (Brown et al., 2020).



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## LANGUAGE MODELS AND CREATIVITY RESEARCH

Language models, particularly when fine-tuned for these purposes, are becoming an important tool in creativity research and specifically in creativity scoring where they surpass accepted methods such as semantic distance in correlation with human ratings (Acar et al., 2023; Luchini et al., 2023; Organisciak, Acar, Dumas, & Berthiaume, 2023). Organisciak et al. (2023) developed a GPT-based LLM called Ocsai. The model was trained and evaluated against one of the largest collections of human-judged AUT responses, with 27 thousand responses collected from nine past studies, and achieved up to $r = .81$ correlation with human raters, greatly surpassing current methods ($r = .12$–$.26$). The authors further showed that learning transfers to new test items and more importantly that the approach is still robust with a small number of training labels. The researchers showed that to achieve a strong performance ($r > .75$) the model had to be fine-tuned on a minimum of 6,000 examples. This is particularly relevant when considering the difficulties and high cost of gathering human-rated responses to creative thinking tasks.

Building on the advancements in English-based LLMs for automated creativity scoring, recent research is extending its scope to non-English LLMs as well. Patterson et al. (2023) utilized two deep-learning-based semantic models to assess verbal creativity in twelve different languages. Their results showcase the potential of these models to align closely with human creativity ratings and demonstrate the adaptability of automated scoring methods for multilingual creativity assessment. Similarly, Goecke et al. (2024) presented a proof-of-concept for the automated scoring of scientific creative thinking tasks in German. Using the XLM-RoBERTa model, they analyzed over 13,000 responses, revealing strong correlations ($r = .81$) between the model's predictions and human ratings. Additionally, Zielińska, Organisciak, Dumas, and Karwowski (2023) showed the efficacy of AI-based scoring models on divergent thinking tasks in Polish, achieving close-to-perfect correlation with human ratings, further validating the reliability of these models even for responses created in different languages.

Language models have also shown success in creative question-asking research (Raz et al., 2024). For example, research done by Raz et al. (2024) introduced a novel approach to automatically score an open-ended questions creativity task, the AQT (Raz et al., 2023), for its Bloom taxonomy complexity using a fine-tuned LLM. Their results reveal that the LLM-generated Bloom scores correlated strongly with human ratings—($r = .77$), greatly exceeding previous baselines. These results highlight the unique ability of models based on LLMs to accurately predict ratings of question-asking tasks, and contribute to linguistic-heavy fields such as creativity.

In addition to being used as scoring tools, there is a bigger debate about whether LLMs can be creative themselves or possibly even more creative than humans (Cropley, 2023; Hubert et al., 2024; Koivisto & Grassini, 2023; Stevenson, Smal, Baas, Grasman, & Maas, 2022). Stevenson et al. (2022) used Open AI's generative LLM, GPT-3, to test whether it can provide creative solutions to one of the most commonly used tests in creativity research—Guilford's alternative uses test (AUT). They then compared its performance to previously collected human responses on expert ratings of originality, usefulness, surprise, flexibility of each set of ideas, and the semantic distance. Their results show that—on the whole—humans currently outperform GPT-3, with human responses rated higher on originality, semantic distance, and surprise. However, GPT-3's responses were rated as more useful. The authors believe it is only a matter of time before GPT-3 or further variants catch up on this particular task (Stevenson et al., 2022). Further work done by Hubert et al. (2024) also aimed to assess the creative potential of humans in comparison with AI. In their study, human participants ($N = 151$) and the newer GPT-4 model provided responses for several divergent thinking creativity tasks—the AUT, consequences task, and divergent associations task. They found that GPT-4 was robustly more creative along each divergent thinking measurement in comparison with the human counterparts. Specifically, when controlling for fluency of responses, AI was more original and elaborate (Hubert et al., 2024).

It is therefore safe to say that LLMs perform at least as well as humans on several benchmark creativity tasks, and thus, further work is needed on elucidating the question of whether these models are truly creative. Perhaps it's time to update the measures we use to define creativity, whether as suggested by Cropley (2023) or by integrating additional measures of creativity such as curiosity and agency as suggested by Brainard (2023). This is further complicated by the fact that while humans have limited recall and memory capabilities, LLMs have potentially endless memory and instantaneous recall abilities, and many creativity tasks may also be in their training data (Gilhooly, 2024) thus contaminating results. While GPT may seem to perform better than humans at verbal divergent thinking tasks, the nature of its performance draws attention to the significant differences between "thinking divergently"—churning out many distantly connected



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ideas and "being creative"—which Brainard (2023) describes so eloquently as being motivated by a desire to attain epistemic goods such as knowledge and understanding, out of a motivation to discover and understand.

## TOWARD A THEORY OF QUESTION-ASKING AND PROMPT-ENGINEERING

In the past couple of years, a new challenge has arisen for human beings: Besides learning to communicate and interact with each other, we need to learn to interact with machines as well. Such interactions may facilitate effective learning where the strategic use of prompts becomes a vital skill in guiding the capabilities of generative AI and optimizing human–machine engagements (Bozkurt & Sharma, 2023). In this section, we suggest that prompt engineering and question-asking are two intricately connected endeavors that share a common foundation of essential skills, inherently related to creativity. Just as master painters skillfully manipulate their brushes on a blank canvas, prompt engineers and question-askers craft the perfect prompts and inquiries for engaging and thought-provoking interactions. This is accomplished by utilizing tools of creativity, precision, and adaptability to retrieve information in an intelligent and insightful way.

### MASTERING INQUIRY: PROMPT AND QUESTION SIMILARITIES

We will first highlight some of the striking parallels in the skills that support both question-asking and prompting (Table 1), illuminating the transformative potential they hold in the diverse areas of learning, discovery, and understanding. This is by far not a complete analysis of similarities and differences related to question-asking and prompt engineering, merely a starting point for future in-depth exploration and analysis.

#### CREATIVITY

Both prompt engineering and question asking require a creative mindset. In prompt engineering, individuals need to design instruction-based prompts or cues that effectively elicit desired responses or actions from the language model. This involves not only a deep understanding of the language model's capabilities and limitations but also an imaginative approach to framing questions or tasks in a way that guides the model toward generating insightful, accurate, and relevant outputs. Similarly, when asking questions, individuals must craft inquiries that provoke thought, encourage exploration, promote meaningful discussions, and above all—that reduce entropy and acquire the desired knowledge (Ivancovsky et al., 2024; Kenett et al., 2023; Raz et al., 2023). In both cases, creativity is essential to formulating engaging and effective prompts or questions.

TABLE 1. Key Similarities Between Question and Prompt Engineering Skills

| Skill | Application in Question Asking | Application in Prompt Engineering |
|---|---|---|
| Creativity | Crafting inquiries that provoke thought and encourage exploration | Designing prompts that elicit desired and insightful responses from AI |
| Clarity and Precision | Articulating thoughts and ideas clearly and concisely | Conveying instructions precisely to minimize misunderstandings |
| Adaptability | Adjusting inquiries based on the audience's knowledge level and interests | Tailoring prompts to the task and language model capabilities |
| Critical Thinking | Stimulating deeper analysis and reflection through strategic questioning | Considering potential outcomes and responses for meaningful interactions |
| Empathy | Formulating respectful, inclusive inquiries that encourage open communication | Optimizing language model responses through empathetic consideration |
| Cognitive Flexibility | Approaching problems from different angles and modifying questions based on the conversation | Iterating with various prompts to optimize results |
| Goal Orientation | Aiming to obtain relevant information or insights for specific issues | Eliciting specific responses that align with the intended purpose |



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## CLARITY AND PRECISION

Effective prompts and questions are required to be precise to convey the intended message or elicit specific information. Ambiguity in prompts can lead to misunderstandings or unproductive responses from the language model (Lin, 2024; Urban et al., 2023), while unclear questions may result in vague answers or confusion. Furthermore, Boyce-Jacino and DeDeo (2020) explore how question opacity can add complexity and affect the difficulty of answering, underscoring the important role of question design and direct cues in communication. The cognitive load theory suggests that individuals' ability to process information is limited, and well-constructed prompts or questions can minimize cognitive load, leading to better understanding and responses (de Jong, 2010). To achieve their objectives, both prompt engineering and question asking rely on the ability to articulate thoughts and ideas clearly and concisely to allow easy processing.

## ADAPTABILITY

Both processes involve adaptability to different contexts and audiences. In prompt engineering, individuals need to tailor prompts to suit the specific task to ensure relevance and fit the capabilities of the language model in use. Similarly, when asking questions, adaptability comes into play as individuals adjust their inquiries based on the knowledge level, interests, and perspectives of the people they are engaging with. As demonstrated by Tian et al. (2023), the strategic use of diverse prompting strategies has been shown to significantly enhance the reasoning abilities of LLMs, underscoring the pivotal role of adaptability in optimizing the outcomes of both prompt engineering and question asking processes.

## CRITICAL THINKING

Effective prompt engineering and question asking both require critical thinking skills. In prompt engineering, the respondent is an artificial intelligence language model, which, despite remarkable progress, remains a statistical model attempting to anticipate human interpretation and responses to such prompts. Thus, individuals must critically consider potential outcomes and responses to ensure meaningful interactions (Lo, 2023). Similarly, in question asking, critical thinking plays a pivotal role in crafting inquiries that stimulate deeper analysis and reflection. Questions, when crafted through the lens of critical thinking, are not merely requests for information but are strategically designed to challenge assumptions, explore underlying concepts, and open avenues for new insights (Albergaria-Almeida, 2011; Santoso, Yuanita, & Erman, 2018). Additionally, individuals must exercise discernment regarding the respondent's background, prior knowledge, and characteristics to critically examine the answers.

## EMPATHY

While possibly surprising to some, both processes can benefit from empathy—the ability to understand and consider others' perspectives and emotions. In prompt engineering, empathetic consideration can enhance interactions by tailoring prompts to optimize the language model's response and ensure a more effective exchange. Lee, Lee, Shin, Bae, and Hahn (2023) have shown how integrating psychotherapeutic principles into prompt engineering significantly enriches the empathetic depth of interactions with LLMs, underlining the pivotal role empathy plays in the development and application of advanced AI communication strategies. In question asking, empathy is crucial in formulating inquiries that are respectful, inclusive, and encourage open communication, reflecting the nuanced understanding that effective questioning often relies on an empathic process, where the inquirer subtly invokes empathy through tone and nonverbal cues to elicit a more engaged and meaningful response (Slote, 2017). Thus, we suggest that empathy helps in facilitating meaningful interactions with both humans and AI language models.

## COGNITIVE FLEXIBILITY

Both processes involve cognitive flexibility to approach problems from different angles and assist the process of information search (Dommes, Chevalier, & Lia, 2011). In prompt engineering, one might need to iterate and experiment with various prompts and examples (as in a few-shot setting) to optimize results. By embracing a broader perspective, developers should draw inspiration from the nuanced characterization of human social interactions, allowing for the cultivation of AI systems that seamlessly integrate with the intricacies of human communication and behavior (Rafner, Beaty, et al., 2023; Rafner, Wang, et al., 2023). In question asking, one should be able to rephrase or modify the questions based on the dynamically evolving conversation. Note that we differ between cognitive flexibility and adaptability in that adaptability refers to



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understanding and adapting to the context of the situation beforehand, while flexibility refers to the process at a later stage.

### GOAL ORIENTATION

Above all, prompt engineering and question asking are goal-oriented activities. In prompt engineering, the goal is to elicit specific responses or actions that align with the intended purpose of the prompts. Similarly, question asking aims to obtain relevant information or insights to address specific issues or promote learning and understanding. It is therefore fair to say that both processes are trying to reduce entropy in order to accomplish some predefined objectives.

It is certainly possible to consider other shared skills, but some obvious questions can already be asked about those mentioned—how can we leverage these shared skills to enhance both prompt engineering and question asking? What cognitive abilities can we nurture and develop to become more adept question-askers or prompt engineers? We will now further explore a range of ideas on how existing theories discussed earlier can be applied to enrich and optimize each of these processes.

## THEORIES IN ACTION: LEVERAGING EXISTING CONCEPTS FOR IMPROVED PROMPTS AND QUESTIONS

As previously discussed, question asking and prompt engineering represent well-established domains with their respective theories. Having established their shared capabilities, we can now shift the focus toward the prospect of cross-application. By exploring the transferability of these theories, we hope to show connections that bring question asking and prompt engineering together, unlocking fresh ways to use these cognitive processes in unison.

In a recent study by Hwang, Challagundla, Alomair, Chen, and Choa (2023), the researchers explored the use of GPT-3.5 for automating the creation and evaluation of multiple-choice questions (MCQs) in chemistry and biology. The methodology involved generating MCQs aligned with Bloom's taxonomy levels and assessing their quality and alignment using the RoBERTa model and item writing flaws criteria, with additional validation by human experts. The findings showed GPT-3.5's capability to produce questions that stimulate higher-order thinking, especially at the "evaluation" level. However, discrepancies between the AI-generated questions and human assessments highlighted the challenges in distinguishing between cognitive levels and in evaluating question quality, particularly at higher Bloom's levels. This study points to the potential of AI in educational assessment while also noting the need for improvements in question generation accuracy and quality assessment at complex cognitive levels.

Hwang et al.'s (2023) study showcases the potential benefits of tailoring prompts according to Bloom's taxonomy by aligning them with each level of Bloom's taxonomy (Table 2). Key terms, which are verbs that fit their respective Bloom level, can provide a starting point for crafting effective prompts. For example, the Bloom level—Remembering, which involves retrieving relevant knowledge, can be described with key terms such as "define," "describe," "label," "list," or "identify." We can then use these terms to craft the following prompt—"List all the planets in the Solar System in order from the Sun."

Thus, human prompt engineers can create more engaging and meaningful interactions with language models, fostering deeper understanding and encouraging higher-order thinking. At the foundational level of Remembering, prompts can be designed to elicit factual information or recall specific details from the language model. By asking for definitions, explanations, or specific data, prompt engineers can assess the model's understanding of basic concepts and facts. Although most language models, and the chatbots based on them, do not excel at factual knowledge as they are trained up to a certain point in time (Brown et al., 2020), it is reasonable to expect that those abilities will improve greatly in the future and that with advances in AI LLMs might become the main source of truth for learners. On the Understanding level, prompts that ask the model to explain a concept in its own words or provide examples to demonstrate comprehension can help gauge the model's ability to grasp and synthesize information, and on the Application level, by presenting language models with context-specific situations and asking them to make predictions or solve practical challenges, prompt engineers can assess the model's capacity to transfer its understanding into relevant applications. Stepping up to the Analysis level, prompts should encourage language models to break down complex information, identify patterns, and make connections between different elements, and for Evaluation we can prompt language models to assess and make judgments about information. Finally, at the Creating level, prompts should integrate ideas or concepts to create something new. Thus, by thoughtfully applying all these levels and their respective key terms to prompt engineering,



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TABLE 2. Example Prompts and Key Terms for LLMs Corresponding to Bloom Taxonomy Levels.

| Bloom's Taxonomy Level | Bloom Level Description | Key Terms | Example Prompt |
|---|---|---|---|
| Remembering | Retrieving relevant knowledge | Define, describe, label, list, identify | "List all the planets in the Solar System in order from the Sun" |
| Understanding | Determining the meaning of instructional messages, including oral, written, and graphic communication | Contrast, generalize, outline, explain, represent | "Explain the significance of the water cycle to Earth's climate system" |
| Applying | Carrying out or using a procedure in a given situation | Apply, examine, interpret, transfer, predict, organize | "Given a dataset of weather conditions, predict whether it will rain tomorrow using a simple linear regression model" |
| Analyzing | Breaking material into its constituent parts and detecting how the parts relate to one another | Discriminate, compare, differentiate, structure | "Compare and contrast the economic impacts of renewable vs. non-renewable energy sources" |
| Evaluating | Making judgments based on criteria and standards | Conclude, appraise, assess, justify, judge, review | "Assess the effectiveness of the recent public health campaign on reducing smoking rates among teenagers" |
| Creating | Putting elements together to form a novel, coherent whole | Compose, design, compile, synthesize, rewrite, formulate | "Design a sustainable city plan that incorporates green energy, efficient public transport, and supports local biodiversity" |

language models can be engaged more deeply, promoting more sophisticated language understanding, and encouraging them to showcase their full capabilities.

Leveraging categorical knowledge can also contribute to enhancing prompt engineering (Table 3). The use of constraint-seeking questions can significantly improve the efficiency of the searching process, by targeting features shared by multiple objects or options. These questions help eliminate several possibilities at each step, and as a result, the language model can narrow down its focus and provide more accurate and relevant responses. In contrast, hypothesis-scanning questions are designed to test specific self-sufficient hypotheses and eliminate only one option at a time. These questions may require more iterations to reach a conclusive answer, as they explore individual possibilities in a more systematic manner. Prompt engineers can leverage hypothesis-scanning questions to encourage the language model to consider various potential solutions and systematically evaluate each option's validity. This approach can be particularly useful when exploring complex problems or generating alternative perspectives (Dua, Gupta, Singh, & Gardner, 2022).

TABLE 3. Example Prompts for LLMs That Leverage Categorical Knowledge

| Question Type | Description | Example Prompt |
|---|---|---|
| Constraint-seeking | Gather broad information by focusing on characteristics common to a category | "Which renewable energy sources are most effective for residential use?" |
| Hypothesis-scanning | Investigating specifics about a single topic or idea within the subject | "Does wind energy produce more electricity compared to solar energy in coastal areas?" |



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Therefore, by utilizing both constraint-seeking and hypothesis-scanning questions in prompt engineering, prompt engineers can strike a balance between efficiency and thoroughness.

On the other hand, while prompt engineering focuses on crafting precise cues to guide language models in generating responses, the principles of few-shot learning from this field can be adapted to construct more meaningful questions when interacting with people. For instance, providing context or a number of examples before asking a question can help the recipient form a general understanding of the topic. This technique not only empowers the respondent to formulate more informed answers but also nurtures their capacity for deeper comprehension and synthesis. Furthermore, few-shot learning in question asking can also be applied to encourage more creative and critical thinking, broadening horizons, and allowing for more diverse answers.

## QUESTION-ASKING, PROMPT-ENGINEERING, AND CO-CREATIVITY

Integrating the skills of question asking and prompt engineering represents a pivotal stride toward co-creativity between humans and artificial intelligence. The necessity to refine prompting strategies through the incorporation of more sophisticated, complex questions is crucial for enhancing the effectiveness of our interactions with AI. By doing so, we not only enhance the capability of language models to generate more precise, relevant, and insightful responses but also elevate the efficiency of human–AI interactions in the future. This convergence of disciplines encourages a richer, more dynamic form of interaction where the boundaries between human creativity and artificial intelligence begin to blur, fostering a shared space that allows for innovation and discovery.

Merging these two lines of research—prompt engineering and question asking—into a unified field of study is essential for unlocking the full potential of human–machine collaboration. It is through this shared effort that we can truly harness the cognitive-like capabilities of language models, pushing beyond simple task completion to the realms of creativity, problem-solving, and learning. By challenging AI with complex inquiries and refining the prompts we use to engage with these systems, we lay the foundation for a future where AI partners with humans in a more meaningful, productive manner and enriches our understanding of cognition, communication, and creativity.

## DISCUSSION

The contribution of this theoretical paper lies in its implications for advancing the understanding of human–machine and human–human interactions and language understanding. By highlighting the shared capabilities of prompt engineering and question asking, our work sheds light on the underlying cognitive mechanisms that both processes entail. Specifically, we aim to bring to the forefront the similarities in how language models process information and how humans engage in meaningful communication through questions. Since LLMs are here to stay, this is critical to enhance such capabilities and facilitate co-creativity and human–machine interaction.

By showcasing how theories from one process can be effectively applied to the other, our work opens possibilities for mutual growth and improvement in both prompt engineering and question asking. The cross-application of theories enhances the strategies used in each process, leading to more efficient and effective interactions and potentially revolutionizing how we engage with intelligent systems and with each other. The implications of such advancements extend beyond the immediate benefits to communication efficiency and effectiveness; they also promise to foster greater interdisciplinary collaboration, bridging gaps between fields such as cognitive science, computational linguistics, and artificial intelligence.

Looking forward, our theoretical paper lays the groundwork for rich future research. Questions remain as to how these insights can be operationalized in practical applications, from educational technologies that adapt to the user's inquiry style to more intuitive and human-like conversational agents in various fields. It also raises crucial questions about ethics, data privacy, and our dependence on technology, urging the development of frameworks to ensure these advancements serve society beneficially.

Building on this foundation, we propose several focused research directions to further explore the connections between question asking, prompt engineering, and creativity. One potential direction involves developing methodologies to improve prompt engineering through the integration of complex question asking techniques. This could involve studying how iterative refinement of prompts in response to LLM feedback mirrors the creative process in humans, thereby enhancing co-creativity between humans and LLMs. Additionally, training programs designed to boost creativity in prompt engineering and question formulation could be developed, empowering individuals to craft more effective and innovative interactions with



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LLMs. Research could also utilize the seven identified similarities between question asking and prompt engineering to devise experiments that assess how these attributes influence the efficacy of prompts and questions in dynamic settings. Moreover, investigating the role of these similarities in iterative processes of creativity and information seeking could provide insights into optimizing few-shot learning scenarios, where examples within a single prompt evolve over the course of interaction. The iterative nature of this interaction reflects a critical aspect of human learning and creativity, which is often characterized by ongoing refinement and adaptation. Such research could lead to more nuanced AI systems that better understand and predict human inquiry patterns, ultimately making technology more intuitive and aligned with human cognitive processes.

Finally, our theoretical alignment between question asking and prompt engineering opens up new lines of inquiry regarding their potential shared cognitive and neural mechanisms. As highlighted above, shared cognitive capacities likely relate to information-seeking behaviors (Ivancovsky et al., 2024; Kenett et al., 2023), such as curiosity, alongside additional traits such as need for cognition (Cacioppo & Petty, 1982) or Openness to Experience (McCrae & Greenberg, 2014). For example, such research can focus on the relation of individual differences in creativity and Internet use, an issue that has so far been shown to lead to counterproductive results (Berker, 2018; Kenett, 2023; Oliva & Storm, 2023). At the neural level, future work can examine how the neural mechanisms that facilitate question asking and prompt engineering relate to recent findings on the neural mechanisms that facilitate cognitive search processes (e.g., Lundin, Brown, Johns, & Todd, 2023; Ovando Tellez et al., 2024, 2022). Thus, the implications of our work invite interdisciplinary efforts to explore and harness the potential of enhanced human–machine interactions for social benefit and additional research.

## CONCLUSION

In conclusion, this theoretical paper underscores the role of curiosity and information-seeking in shaping the fields of question asking and prompt engineering. By drawing on diverse theoretical frameworks, such as Bloom's Taxonomy for question types and techniques in few-shot and zero-shot learning for prompt engineering, we highlight how these disciplines not only complement but also enhance each other. The identified similarities between these fields and the skills they require, including creativity which strongly connects them, illustrate the underlying cognitive processes that drive both human and machine interactions.

Our theoretical exploration shows that effectively combining these approaches can lead to more sophisticated, intuitive, and effective communication tools. These insights pave the way for future research that seeks to optimize the interplay between human cognitive abilities and machine learning capabilities, and boost creativity in human interactions. As we advance, it is crucial to continue bridging the gap between computational linguistics, cognitive science, and artificial intelligence, while constantly pushing the boundaries of creative communication and interactive technologies. Similarly to the significance of asking more complex questions in creativity (Raz et al., 2023), we should strive to create more complex and tailored prompts to enhance human–machine interactions and acts of co-creativity.

## CONFLICT OF INTEREST

The authors declare no conflict of interests.

## DATA AVAILABILITY STATEMENT

No data were used for this paper.

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Gal Sasson Lazovsky, Tuval Raz, Yoed N. Kenett, Technion – Israel Institute of Technology

*Correspondence concerning this article should be addressed to Gal Sasson Lazovsky, Faculty of Data and Decision Sciences, Technion—Israel Institute of Technology, Kiryat Hatechnion, Haifa 3200003, Israel. E-mail: galsasson@campus.technion.ac.il

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